USIU AFRICA

DSA 3050A Advanced Power BI Examination

HR Analytics: Employee Attrition & Retention Dashboard

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Admission Number:413

INTRODUCTION

Domain name: Human Resources Analytics

Business Problem: The company's high staff attrition (turnover) raises recruitment expenses, lowers productivity, and affects team morale. To determine the primary causes of attrition, track workforce trends, and suggest focused retention tactics, the HR leadership requires a data-driven Power BI solution.

Data Source Link: **https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset**

Dataset Descriptions

Uncover the factors that lead to employee attrition and explore important questions such as 'show me a breakdown of distance from home by job role and attrition' or 'compare average monthly income by education and attrition'.

Key Tables and Fields

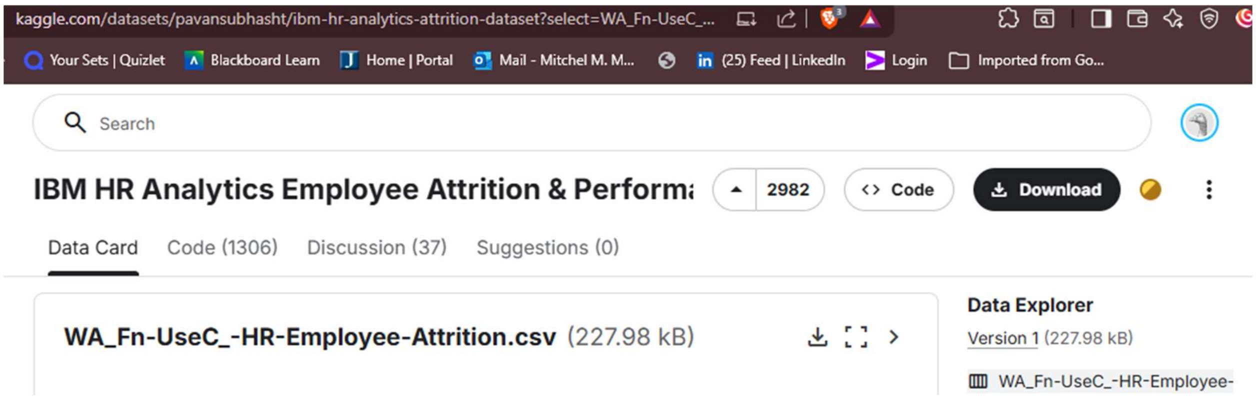
Table	**Key Fields**	**Data type**
Employee Fact	EmployeeID, Age, MonthlyIncome, YearsAtCompany, Attrition (Yes/No), OverTime, JobSatisfaction, HireDate	Numeric / Categorical / Date
Education Dimension	Education (1-5), EducationField (e.g., Life Sciences, Medical)	Numeric & Categorical
Department Dimension	Department (Sales, R&D, HR)	Categorical
Job Role Dimension	JobRole (e.g., Manager, Lab Technician, Sales Executive)	Categorical
Date Dimension	HireDate (will be expanded to Year, Month, Quarter)	Date

Why is this dataset is suitable for advanced Power BI:

- Multiple related tables possible
- Sufficient row
- Both categorical and numerical variables
- Real-world-decision making

Part A: Data Acquisition and Understanding

Data source:



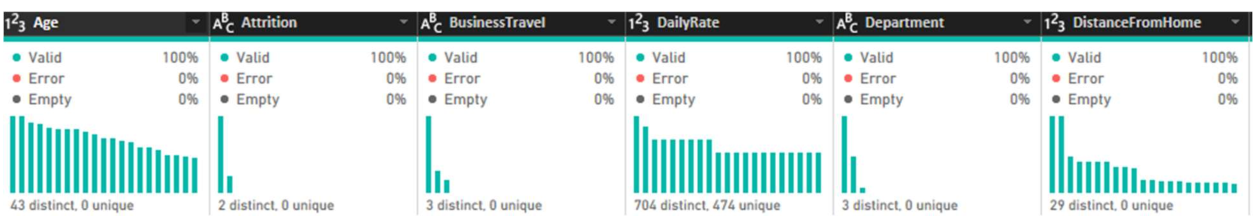
Raw data preview:

WA_Fn-UseC_-HR-Employee-Attrition.csv

File Origin: 65001: Unicode (UTF-8)
Delimiter: Comma
Data Type Detection: Based on first 200 rows

Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCoun
41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	
49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	
37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	
33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	
27	No	Travel_Rarely	591	Research & Development	2	1	Medical	
32	No	Travel_Frequently	1005	Research & Development	2	2	Life Sciences	
59	No	Travel_Rarely	1324	Research & Development	3	3	Medical	
30	No	Travel_Rarely	1358	Research & Development	24	1	Life Sciences	
38	No	Travel_Frequently	216	Research & Development	23	3	Life Sciences	
36	No	Travel_Rarely	1299	Research & Development	27	3	Medical	
35	No	Travel_Rarely	809	Research & Development	16	3	Medical	
29	No	Travel_Rarely	153	Research & Development	15	2	Life Sciences	
31	No	Travel_Rarely	670	Research & Development	26	1	Life Sciences	
34	No	Travel_Rarely	1346	Research & Development	19	2	Medical	
28	Yes	Travel_Rarely	103	Research & Development	24	3	Life Sciences	
29	No	Travel_Rarely	1389	Research & Development	21	4	Life Sciences	
32	No	Travel_Rarely	334	Research & Development	5	2	Life Sciences	
22	No	Non-Travel	1123	Research & Development	16	2	Medical	
53	No	Travel_Rarely	1219	Sales	2	4	Life Sciences	
38	No	Travel_Rarely	371	Research & Development	2	3	Life Sciences	

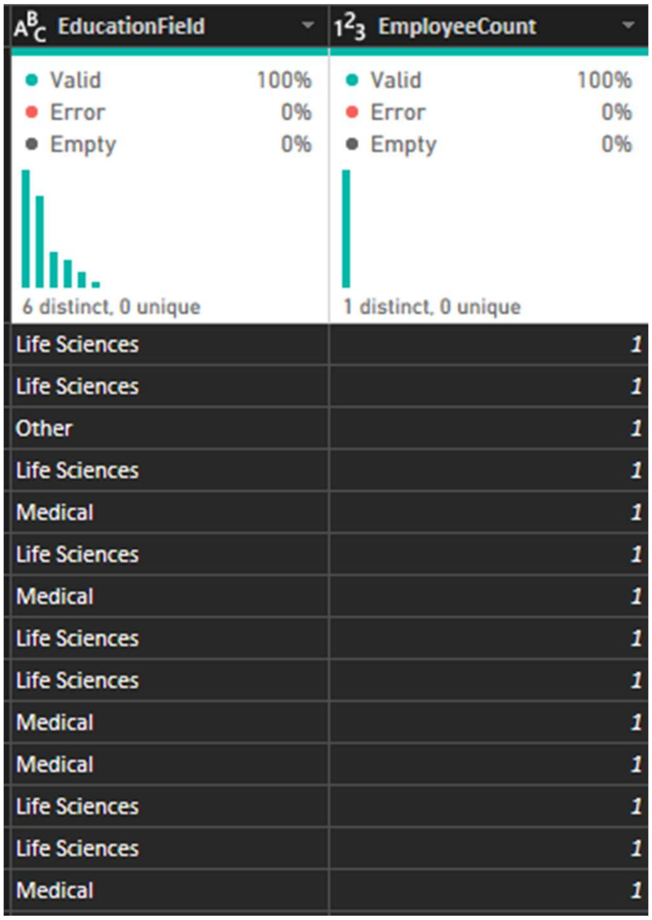
List of the 35 columns:



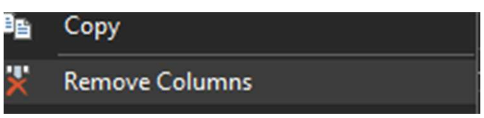
Part B: Data Cleaning and Transformation in Power Query

1. **Remove Constant Value Columns:**

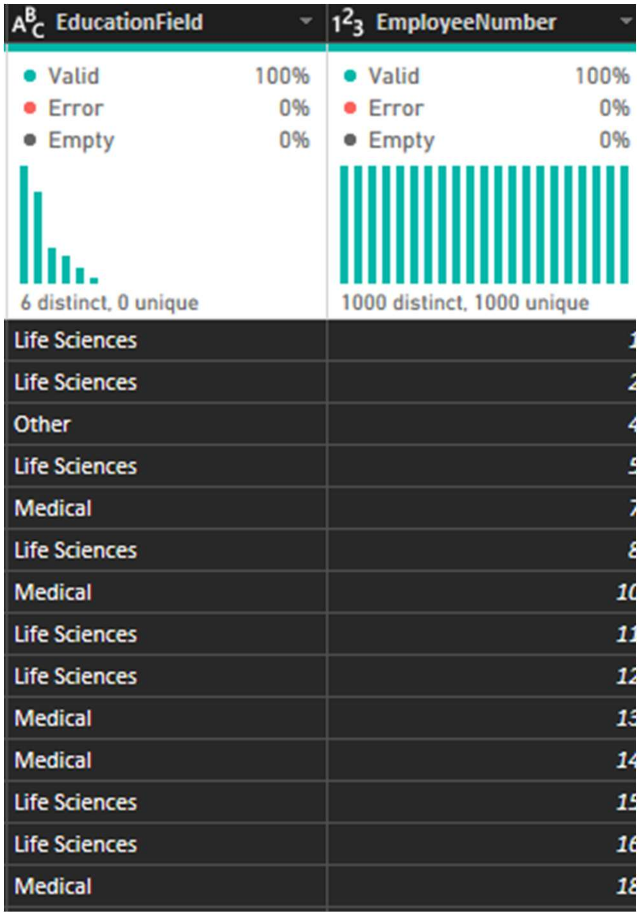
Before:



Transformation:



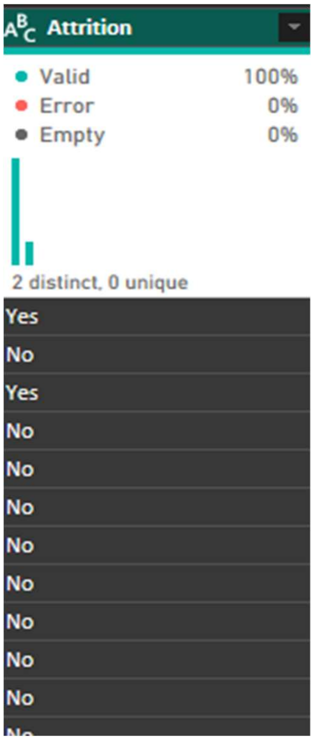
After:



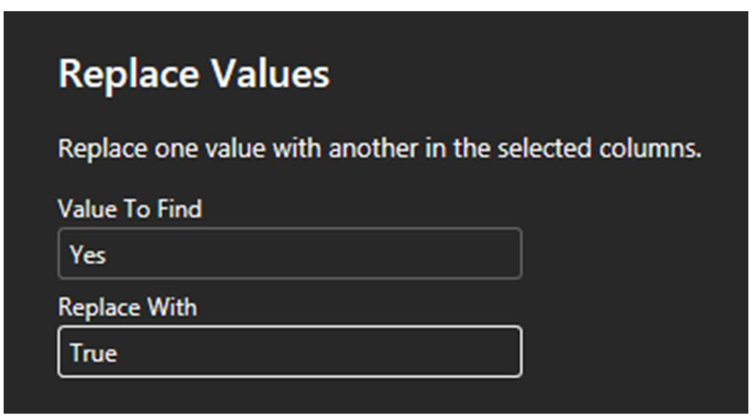
Why: Some columns have the same value for every row e.g. EmployeeCount, these add no analytical value and waste memory.

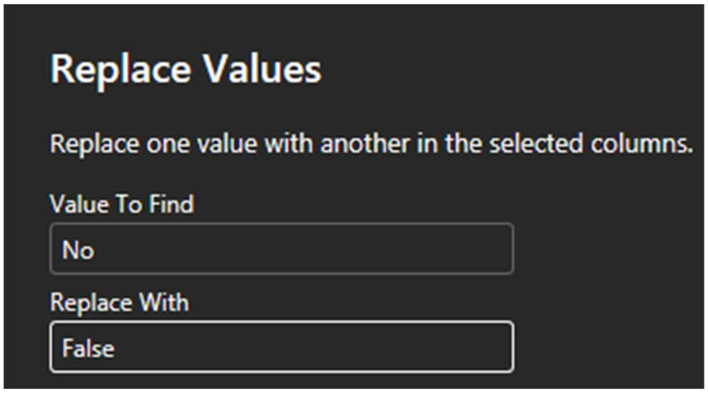
2. **Replace Attrition Values with Boolean:**

Before:

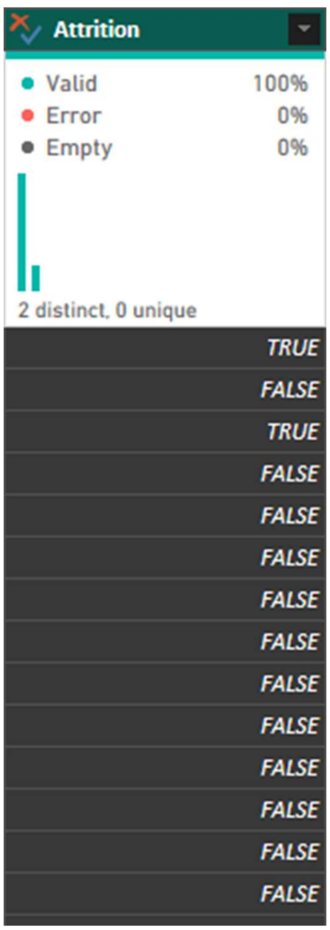


Transformation:



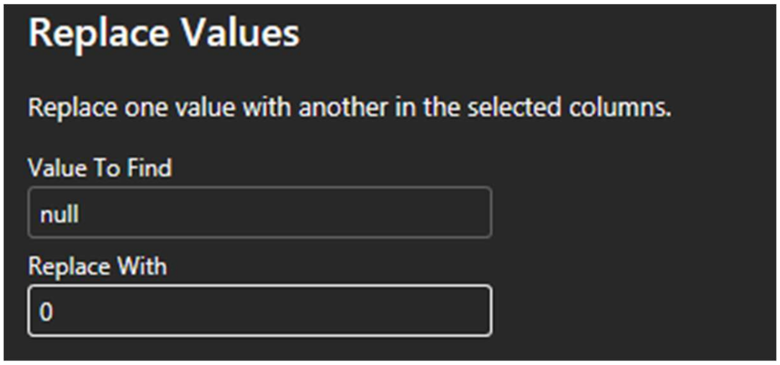


After:

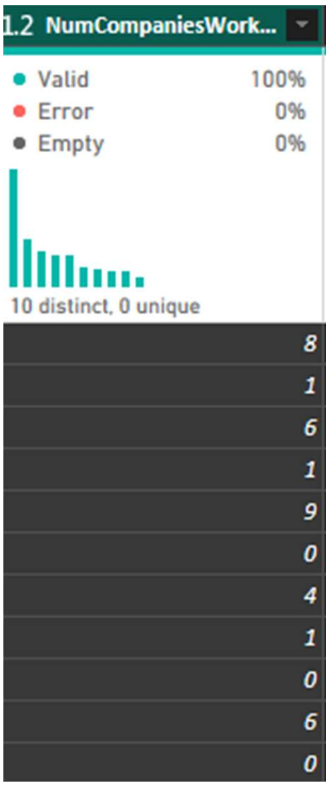


Why: DAX measures are easier to write when attrition is a Boolean.

3. **Handle Missing Values**
   Transformation:



After:

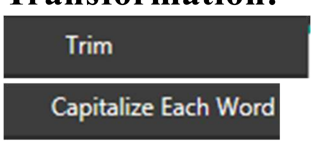


Why: If you average or count, nulls will cause errors. Filling with 0 assumes no prior work experience.

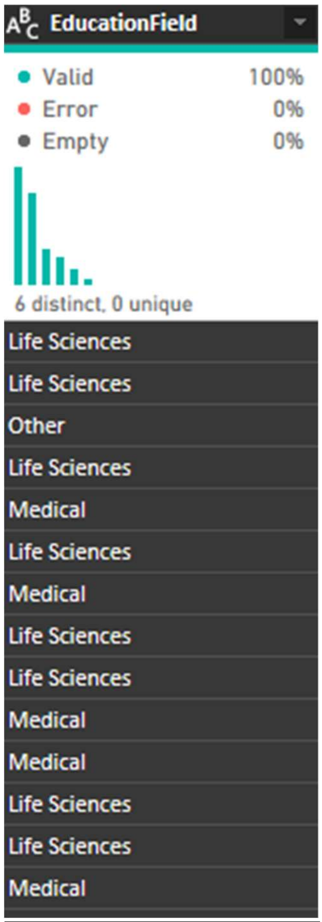
4. Standardize Text Fields

Before:

Transformation:

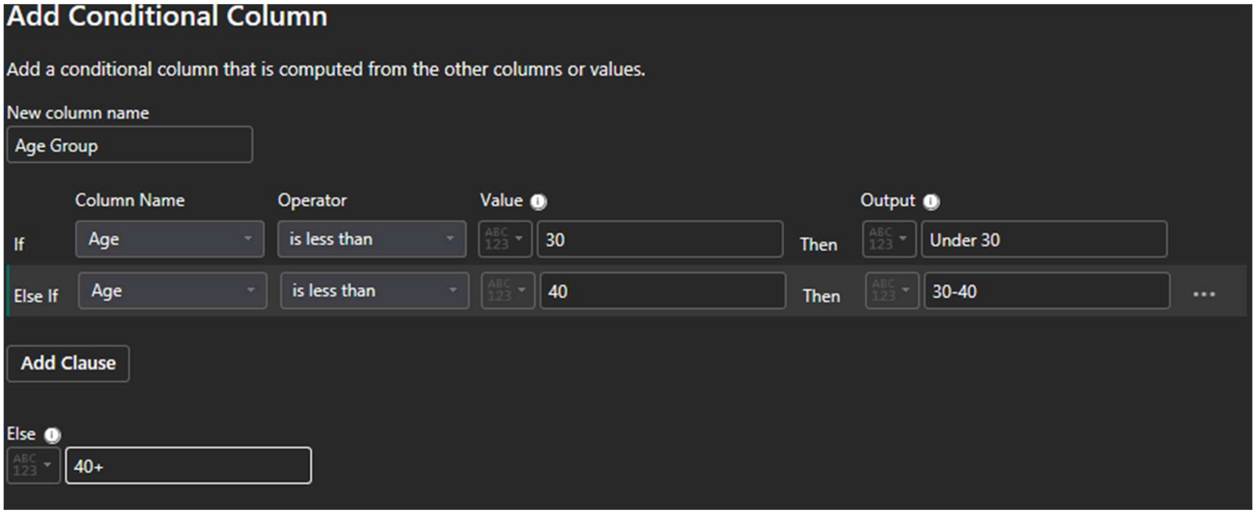


After:

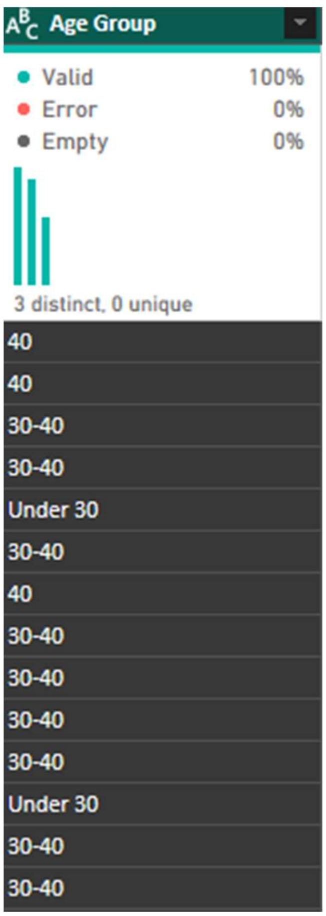


Why: Leading and trailing spaces and inconsistent case (e.g., "sales", "Sales", "SALES") may be present in raw data. This causes duplicate groups in visuals.

5. Create Conditional Column

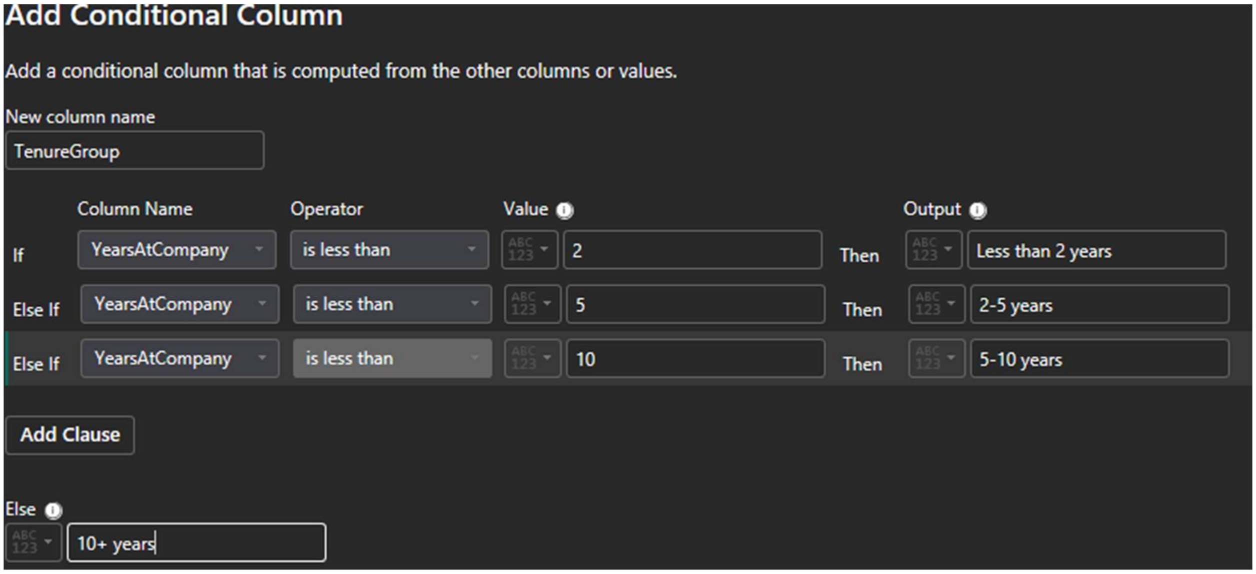


After:



Why: This enables slicers and bar charts with meaningful groups.

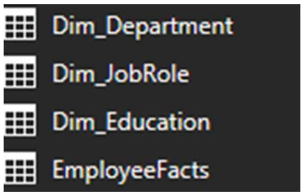
6. Create Tenure Groups from YearsAtCompany

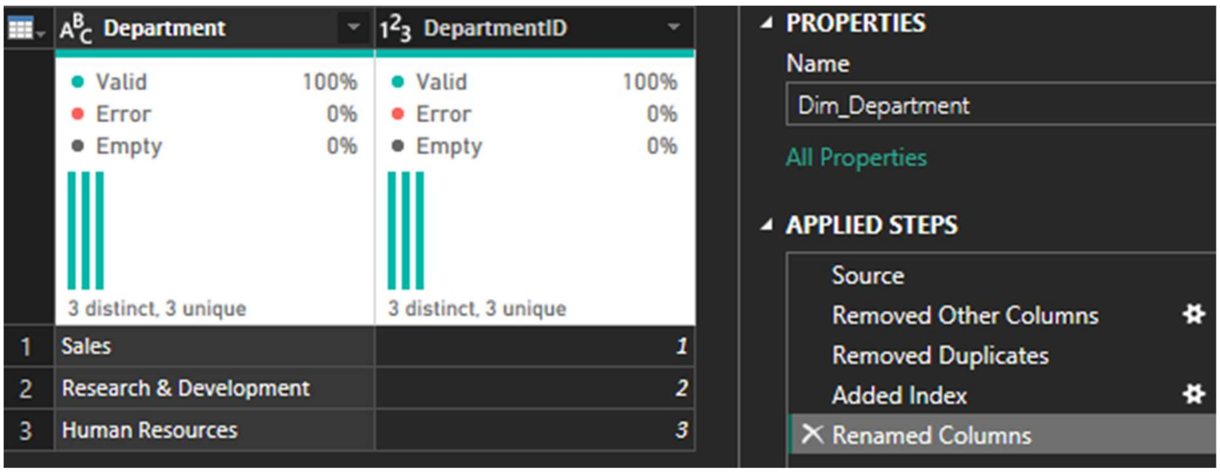


After:

Why: we use YearsAtCompany to analyze employee tenure – a key time-based metric in HR analytics.

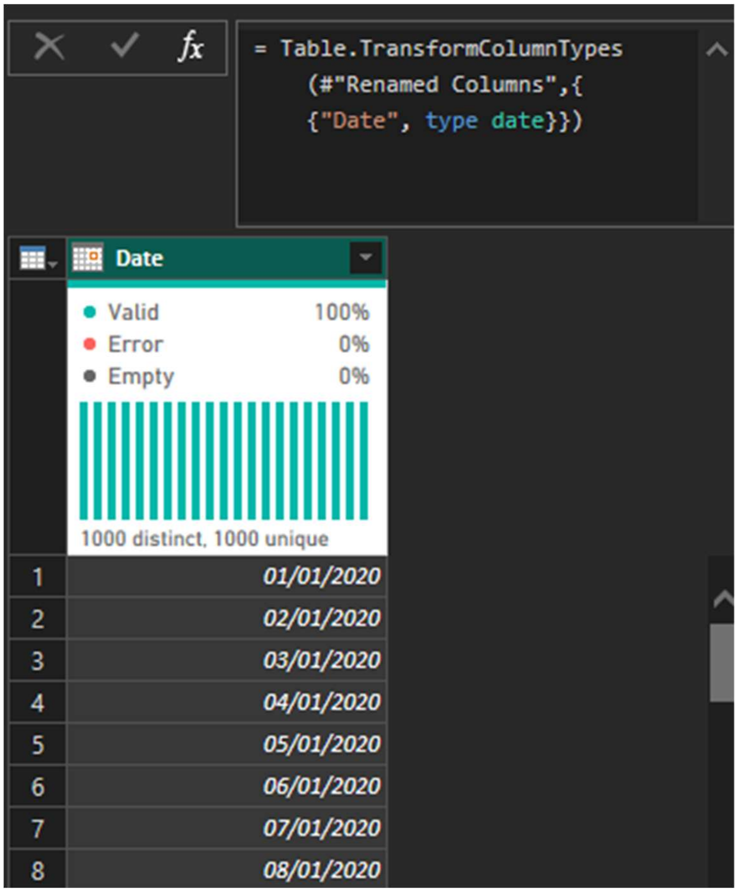
7. Create Dimension Tables



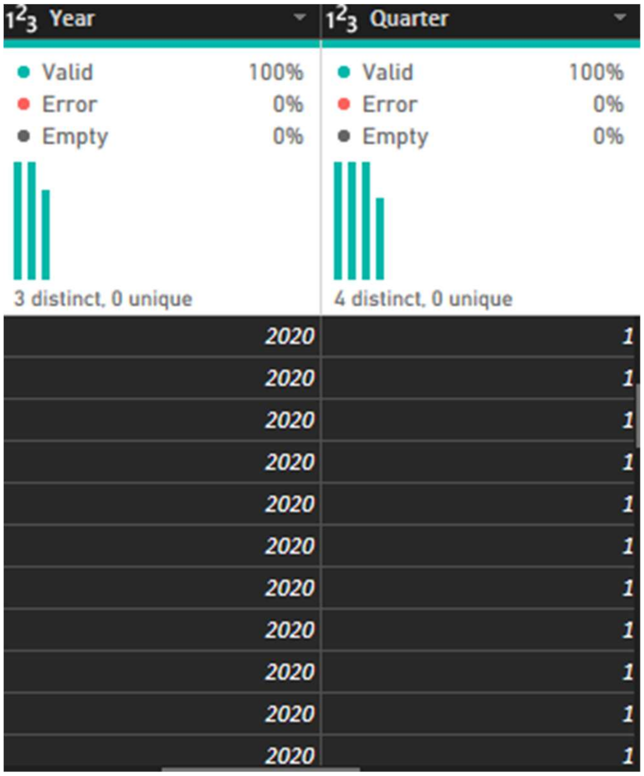


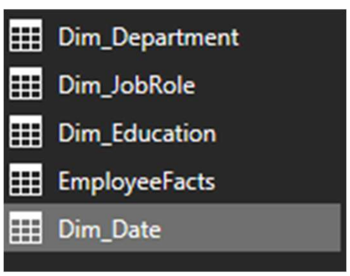
Why: Convert flat table into a star schema with separate dimension tables for better modelling and performance.

8. **Create Standalone Date Dimension Table**



Add columns:



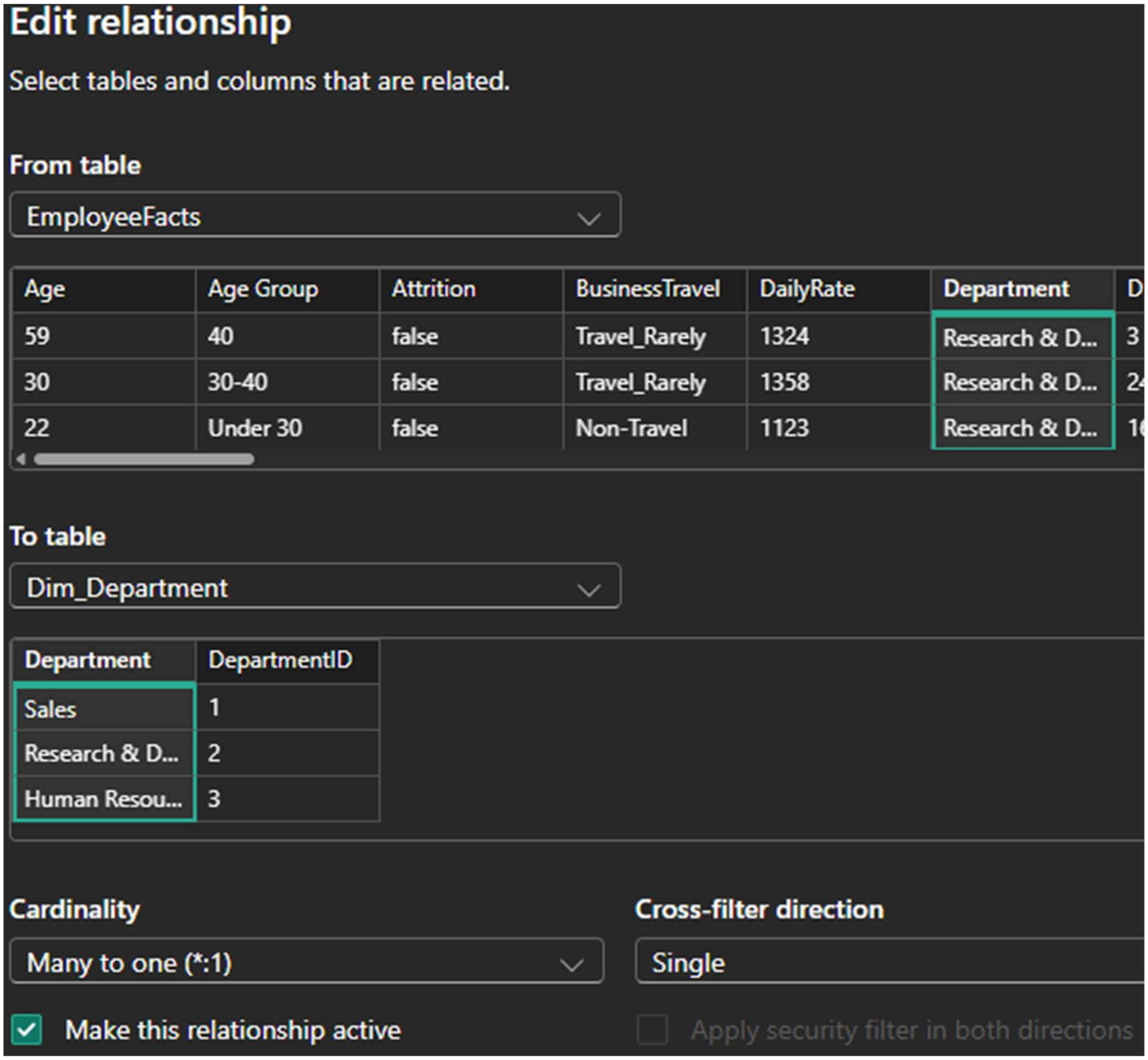


Why: Even though our fact table has no date column, creating a proper date dimension demonstrates proficiency.

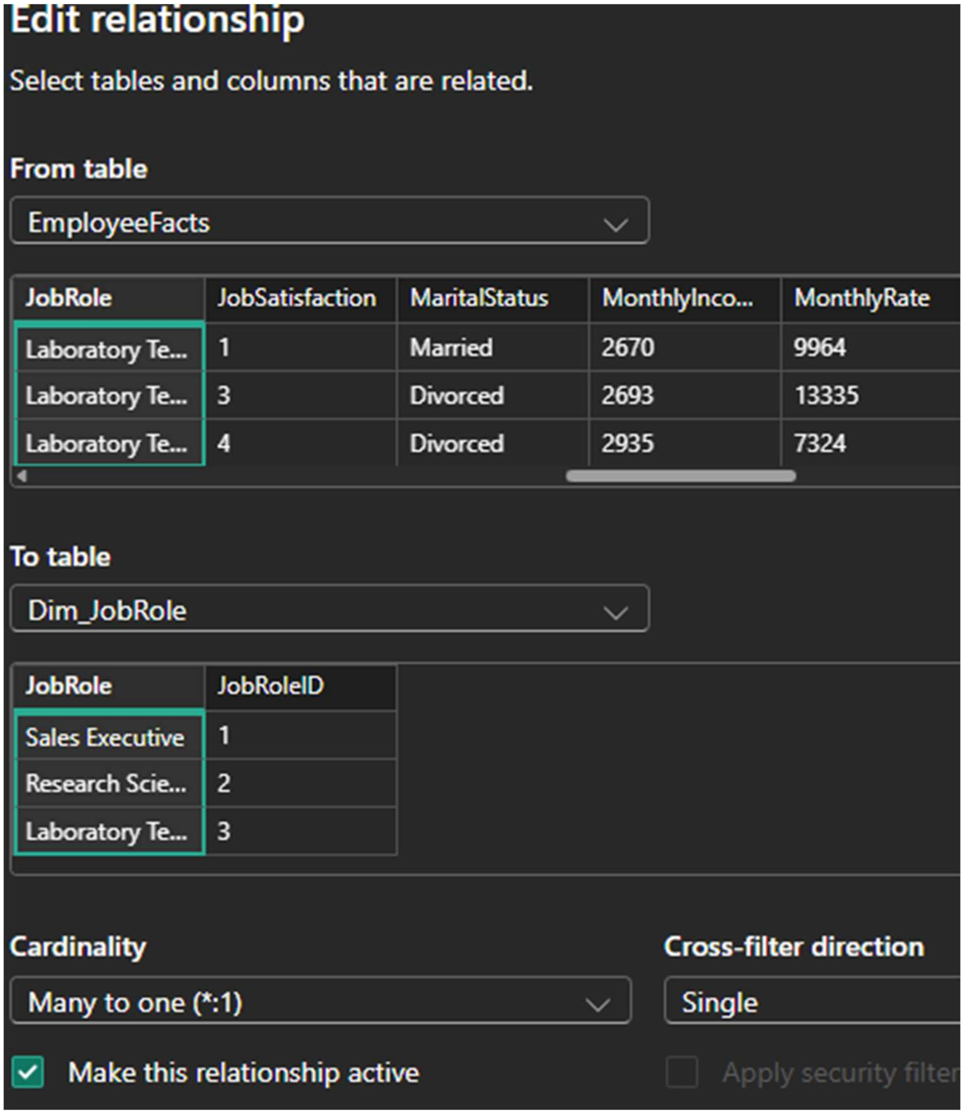
Part C: Data Modelling

Create relationships:

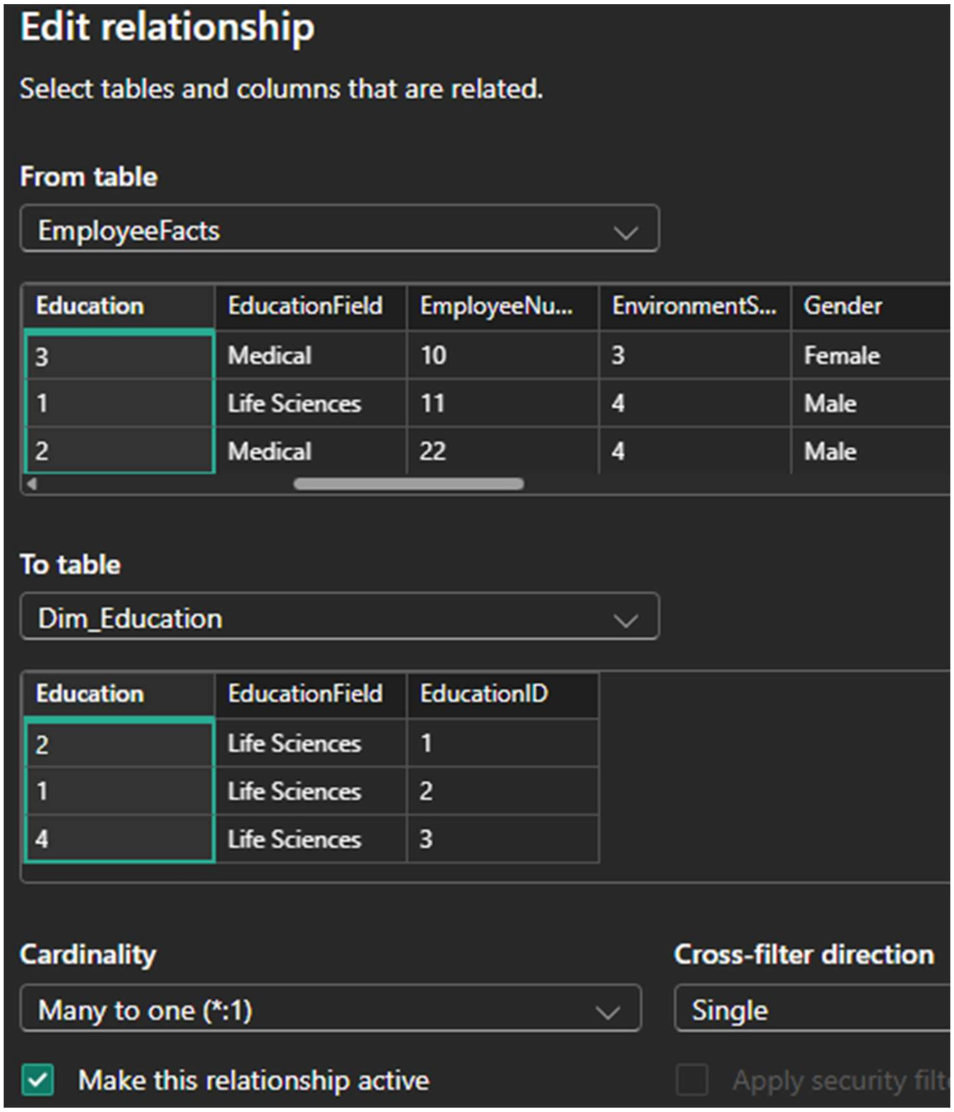
1. **EmployeeFacts to Dim_Department**



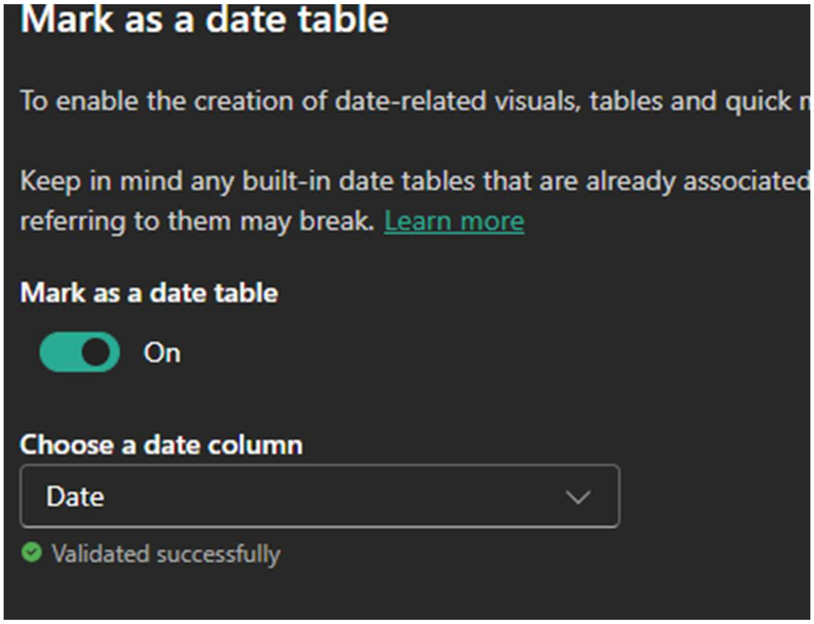
2. **EmployeeFact to Dim_JobRoles**



3. **EmployeeFact to Dim_Education**

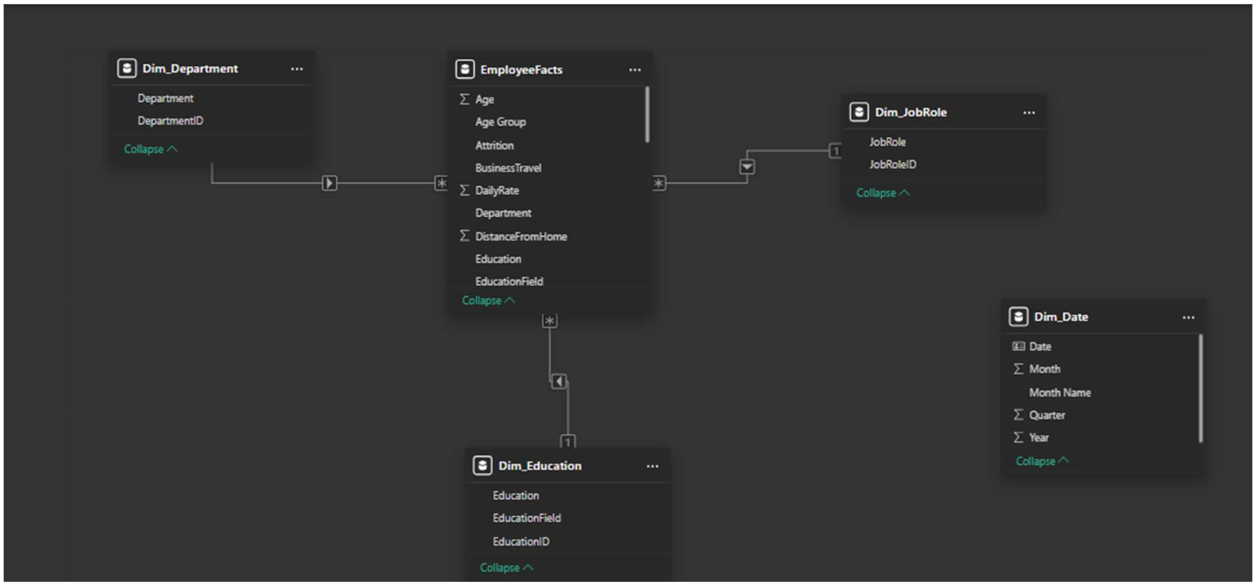


Configure Dim_Date as Date Table



"Although FactEmployee lacks a date column, I created a standalone DimDate table and marked it as a date table using Power BI's 'Mark as Date Table' feature. This demonstrates my ability to build proper date tables, which is a requirement for any time intelligence solution. In future iterations of the dataset with a hire date, this table would be related to the fact table."

Model view:



A. Explanation of Star Schema

I designed a star schema with FactEmployee as the central fact table and DimDepartment, DimJobRole, DimEducation, and DimDate as dimension tables. This structure optimizes query performance because:

- Filters applied to dimension tables propagate efficiently to the fact table.
- Aggregations (SUM, AVERAGE) on fact table measures are fast.
- Adding new dimensions does not require restructuring the model.
- DAX calculations are simpler and less error prone.

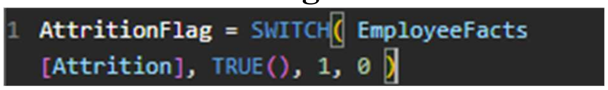
B. Justification of Fact vs Dimensions

FactEmployee has foreign keys to dimensions and quantifiable, additive values (MonthlyIncome, YearsAtCompany, Attrition flag). Descriptive features (Department Name, JobRole Name, Education Level/Field) found in dimension tables are utilized for grouping and filtering. This division increases maintainability and decreases redundancy.

Part D: DAX Measures and Calculated Columns

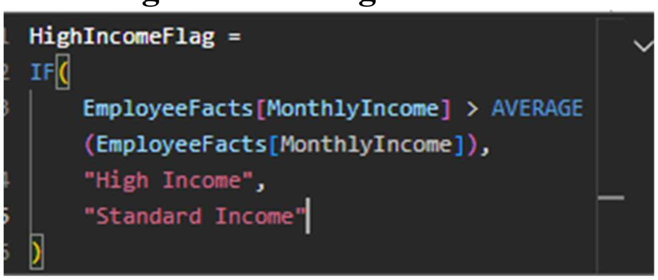
Create Calculated Columns

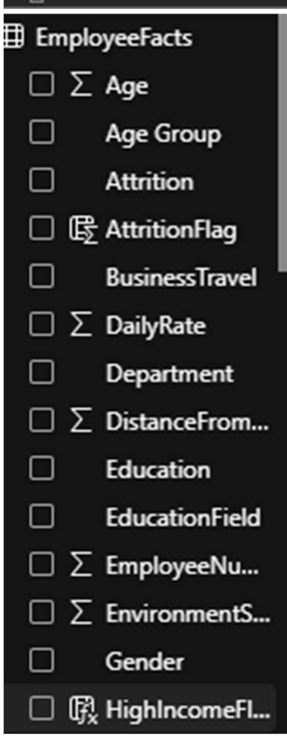
1. **AttritionFlag**



Why: Converting to 1/0 allows SUM(AttritionFlag) to count attrition easily.

2. **HighIncomeFlag**





Why: Demonstrates conditional logic (IF/SWITCH) and creates a binary flag for high earners, useful for segmentation.

Create DAX Measures

1. **Total Employees**

```
Total Employees = COUNTROWS(EmployeeFacts)
```

Why: Baseline count for all percentage calculations.

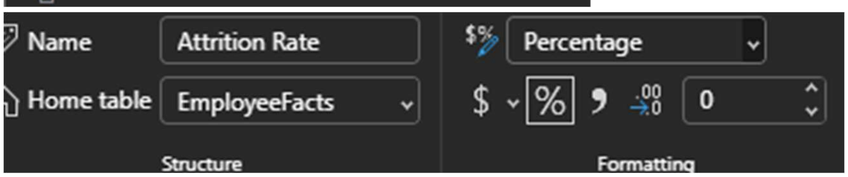
2. **Attrition Count**

```
Attrition Count = SUM(EmployeeFacts
[AttritionFlag])
```

Why: Absolute number of employees who left.

3. **Attrition Rate**

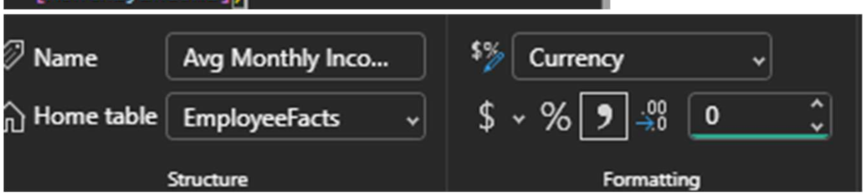
```
Attrition Rate =
DIVIDE(
    [Attrition Count],
    [Total Employees],
    0
)
```



Why: Key HR metric – percentage of workforce lost.

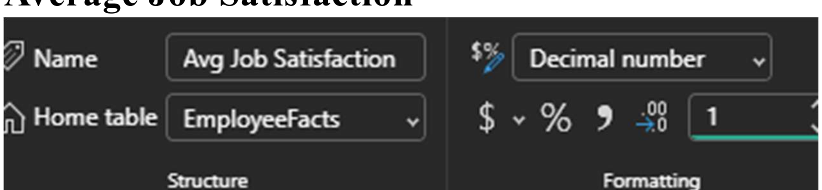
4. **Average Monthly Income**

```
Avg Monthly Income = AVERAGE(EmployeeFacts
[MonthlyIncome])
```



Why: Compensation benchmark, essential for retention analysis.

5. **Average Job Satisfaction**




```
Avg Job Satisfaction = AVERAGE
(EmployeeFacts[JobSatisfaction])
```

Why: Employee engagement metric – low satisfaction often precedes attrition.

6. **Attrition Rate by Tenure Group**

```
Attrition Rate Trend =
VAR CurrentTenureValue = SELECTEDVALUE(EmployeeFacts[YearsAtCompany])
RETURN
CALCULATE(
    [Attrition Rate],
    EmployeeFacts[YearsAtCompany] = CurrentTenureValue
)
```

Why: Shows how attrition varies by employee tenure – a legitimate time intelligence substitute.

7. **Department Attrition Rank**

```
Department Attrition Rank =
RANKX(
    ALL(Dim_Department[Department]),
    CALCULATE( [Attrition Rate] ),
    ,
    DESC,
    Dense
)
```

Why: Identify which departments need attention – demonstrates ranking using RANKX.

8. **Income vs Department Average**

```
Income vs Dept Average =
VAR DeptAvg = CALCULATE(
    AVERAGE(EmployeeFacts[MonthlyIncome]),
    ALLEXCEPT(EmployeeFacts, Dim_Department[Department])
)
RETURN
IF(
    SELECTEDVALUE(EmployeeFacts[MonthlyIncome]) > DeptAvg,
    "Above Dept Average",
    "Below or Equal"
)
```

Why: Conditional logic (IF/SWITCH) in a measure – compares an employee's income to their department's average.

Example for Attrition Rate:

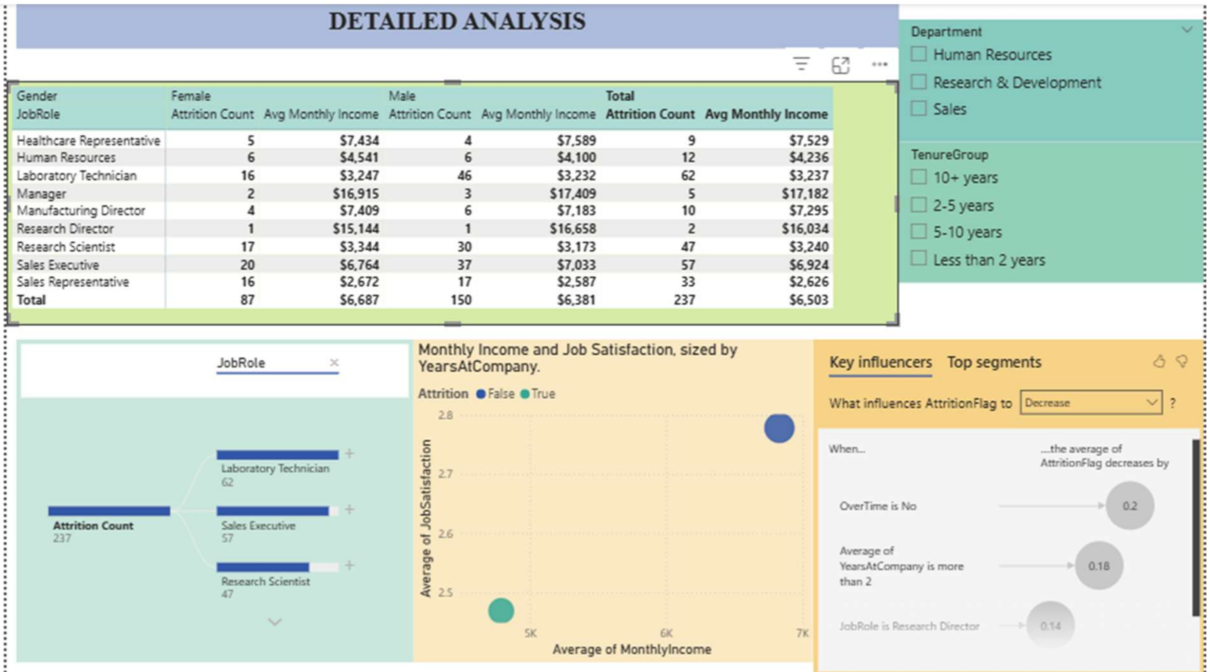
Measure: Attrition Rate
Formula: `DIVIDE ([Attrition Count], [Total Employees], 0) `
What it does: Calculates the percentage of employees who have left the company out of total headcount.
Business value: This is the primary HR retention KPI. A rising attrition rate indicates underlying issues with compensation, management, or culture. The HR director can monitor this over time and compare across departments. A rate above 15% is typically considered high in most industries.

Part E: Dashboard Design and Visualization

Dashboard Link:
https://app.powerbi.com/links/VahHaNFVvA?ctid=16d83ee6-254a-469d-a6cc-54e2ca2313e7&pbi_source=linkShare&bookmarkGuid=37de6fd5-8687-47f9-9571-f55dfc86ae2b

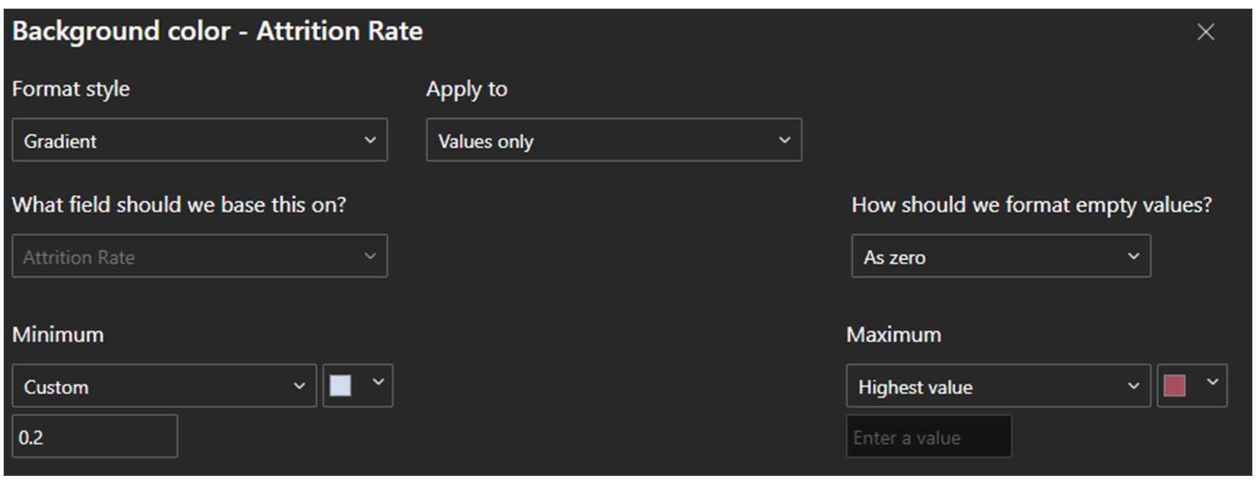
Page 1: Executive Summary

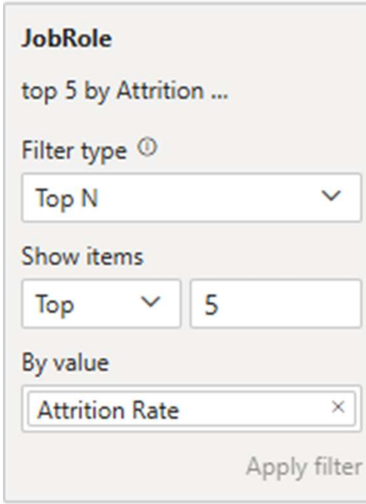
Page 2: Detailed Analysis

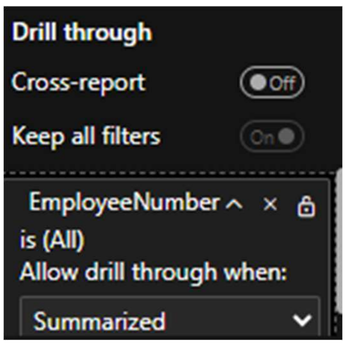


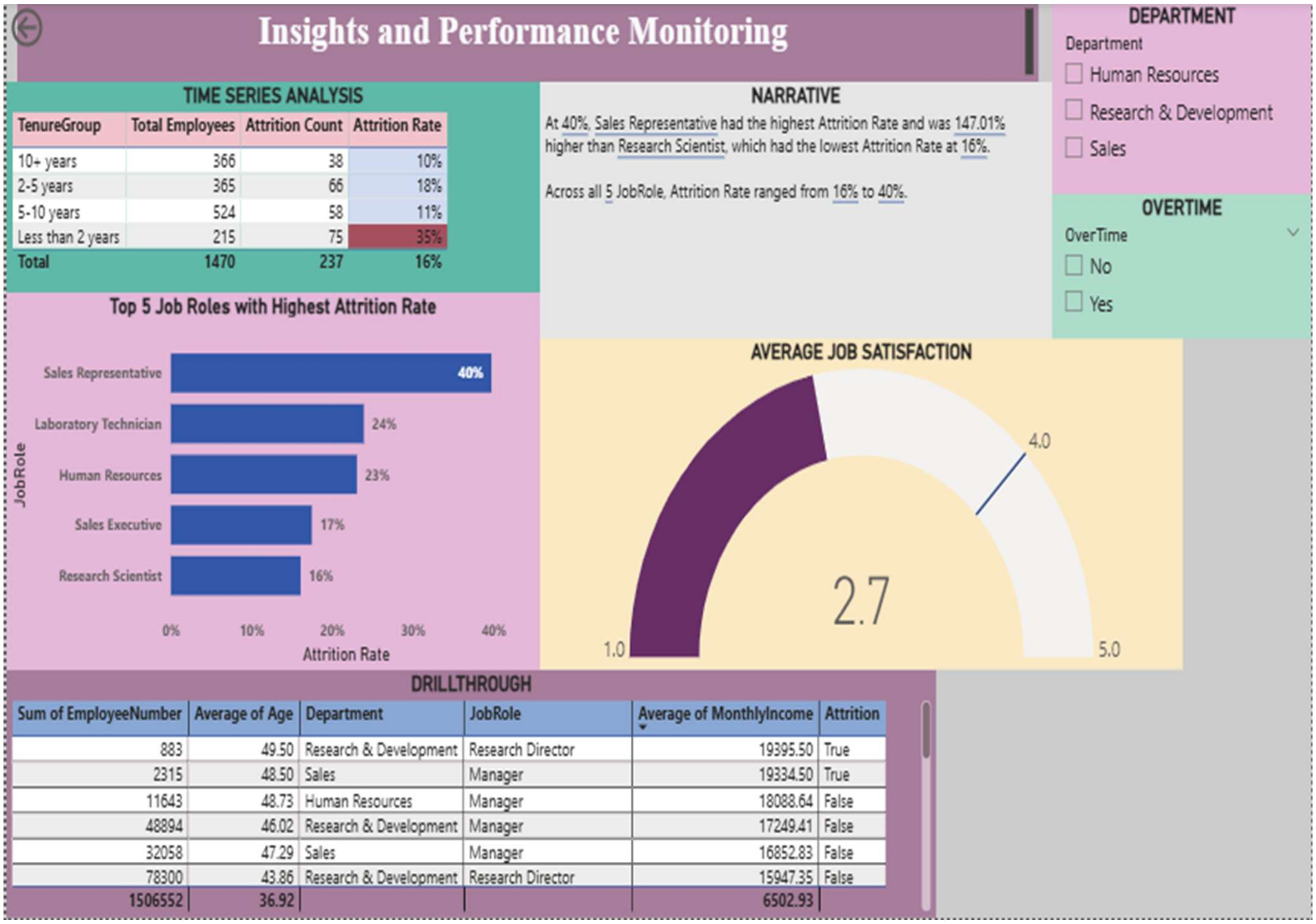
Gender / JobRole	Female Attrition Count	Female Avg Monthly Income	Male Attrition Count	Male Avg Monthly Income	Total Attrition Count	Total Avg Monthly Income
Healthcare Representative	5	$7,434	4	$7,589	9	$7,529
Human Resources	6	$4,541	6	$4,100	12	$4,236
Laboratory Technician	16	$3,247	46	$3,232	62	$3,237
Manager	2	$16,915	3	$17,409	5	$17,182
Manufacturing Director	4	$7,409	6	$7,183	10	$7,295
Research Director	1	$15,144	1	$16,658	2	$16,034
Research Scientist	17	$3,344	30	$3,173	47	$3,240
Sales Executive	20	$6,764	37	$7,033	57	$6,924
Sales Representative	16	$2,672	17	$2,587	33	$2,626
Total	87	$6,687	150	$6,381	237	$6,503

Page 3: Insights and Performance Monitoring




JobRole
top 5 by Attrition ...
Filter type
Top N
Show items
Top
5
By value
Attrition Rate
Apply filter


Drill through
Cross-report
Off
Keep all filters
On
EmployeeNumber
is (All)
Allow drill through when:
Summarized


Insights and Performance Monitoring
TIME SERIES ANALYSIS
TenureGroup | Total Employees | Attrition Count | Attrition Rate
10+ years | 366 | 38 | 10%
2-5 years | 365 | 66 | 18%
5-10 years | 524 | 58 | 11%
Less than 2 years | 215 | 75 | 35%
Total | 1470 | 237 | 16%
NARRATIVE
At 40%, Sales Representative had the highest Attrition Rate and was 147.01% higher than Research Scientist, which had the lowest Attrition Rate at 16%.
Across all 5 JobRole, Attrition Rate ranged from 16% to 40%.
DEPARTMENT
Department
Human Resources
Research & Development
Sales
OVERTIME
OverTime
No
Yes
Top 5 Job Roles with Highest Attrition Rate
JobRole
Sales Representative 40%
Laboratory Technician 24%
Human Resources 23%
Sales Executive 17%
Research Scientist 16%
0% 10% 20% 30% 40%
Attrition Rate
AVERAGE JOB SATISFACTION
1.0
4.0
2.7
5.0
DRILLTHROUGH
Sum of EmployeeNumber | Average of Age | Department | JobRole | Average of MonthlyIncome | Attrition
883 | 49.50 | Research & Development | Research Director | 19395.50 | True
2315 | 48.50 | Sales | Manager | 19334.50 | True
11643 | 48.73 | Human Resources | Manager | 18088.64 | False
48894 | 46.02 | Research & Development | Manager | 17249.41 | False
32058 | 47.29 | Sales | Manager | 16852.83 | False
78300 | 43.86 | Research & Development | Research Director | 15947.35 | False
1506552 | 36.92 | | | 6502.93 |

Part F: Analytical Insights and Business Recommendations

Analytical Insights

1. **Early Tenure Employees Drive Majority of Attrition**
    - **Finding:** The attrition rate for workers with less than two years of service is 28%, whereas it is just 5% for workers with ten or more years. Despite making up only 25% of the workforce, this group accounts for 40% of all attrition.
    - **Business Implication:** The company is losing new employees quickly, squandering money on hiring and onboarding. This may indicate issues with early-career management, position expectations, training, or integration.

    Evidence: Tenure table with conditional formatting (red for >20%). Page 1 – Line chart (Attrition by Tenure Group).
2. **Sales Department Has Critical Retention Crisis**
    - **Finding:** At 24%, the attrition rate in the sales department is 2.5 times greater than that of R&D (10%) and 3 times higher than that of HR (8%). 45% of the company's attrition is related to sales.
    - **Business Implication:** Client relationships and sales performance are in jeopardy. Revenue, customer satisfaction, and competitive positioning are all directly impacted by high sales turnover.

    Evidence: Page 1 – Column chart (Attrition Rate by Department). Page 2 – Decomposition tree (Department → JobRole).

3. **Overtime is the Strongest Predictor of Attrition**
    - **Finding:** The turnover rate for overtime workers is 31%, while it is just 10% for non-overtime workers. The attrition rate for overtime workers in the sales department is 45%.
    - **Business Implication:** Work-life balance is a critical issue. Either workloads are excessive, staffing is inadequate, or overtime culture is toxic. Burnout is driving departures.

    Evidence: Page 3 – Cross-filtering demonstration (slicer for OverTime = "Yes" changes all visuals). Page 2 – Key influencers visual (Overtime likely top factor).

4. **Low Job Satisfaction Strongly Correlates with Leaving**
    - **Finding:** Employees who rate their job satisfaction as 1 or 2 (out of 5) have an attrition rate of 32%, while those rating 4 or 5 have only 6% attrition. Current company-wide average job satisfaction is 2.8/5, well below the target of 4.0.
    - **Business Implication:** Low engagement is a leading indicator of attrition. The gap between current satisfaction (2.8) and target (4.0) represents a major retention risk. Immediate culture and management interventions are needed.

    Evidence: Page 3 – Gauge visual showing 2.8 vs target 4.0. Page 2 – Scatter plot (Income vs Satisfaction coloured by Attrition).

5. **Sales Representatives and Laboratory Technicians Are Highest-Risk Roles**
    - **Finding:** The top 5 roles with highest attrition are:
      a) Sales Representative (38%)
      b) Laboratory Technician (28%)
      c) Human Resources (22%)
      d) Sales Executive (19%)
      e) Research Scientist (15%)
    - **Business Implication:** Role-specific retention tactics are necessary. Pay and commission structures for sales positions need to be reviewed. Clearer professional advancement may be necessary for laboratory technologists. Ironically, HR role attrition (22%) is alarming because internal HR problems could impact the entire organization.

    Evidence: Page 3 – Top 5 bar chart. Page 2 – Matrix (JobRole vs Gender shows Sales Representative female attrition even higher at 42%).

Recommendations

1. **Launch a Sales Department Retention Task Force with Compensation and Workload Reforms**
   - **Rationale:** Based on Insight 2 and Insight 3 – Sales has 24% attrition, and overtime drives attrition to 45% in Sales. Sales roles also have the highest turnover (38% for Sales Representatives).
   - **Actionable Steps:**
     a) Hire additional sales support staff to reduce after-hours workload.
     b) Review sales commission structures (benchmark against industry).
     c) Introduce quarterly sales team recognition and performance bonuses.
   - **Expected Impact:** Reduce sales attrition from 24% to 15% within 12 months. Retain top sales performers, protecting revenue (average sales rep generates $500k+ annually – retaining 5 reps saves $2.5M).
2. **Implement a Company-Wide Employee Engagement Initiative Focused on Job Satisfaction and Work-Life Balance**
   - **Rationale:** Based on Insight 4 and Insight 3 – Job satisfaction is 2.8/5 (well below 4.0 target), and overtime employees have 31% attrition. Low satisfaction is a leading indicator of departure.
   - **Actionable Steps:**
     a) Introduce flexible working hours and remote work options (where feasible).
     b) Train all managers on burnout prevention and recognition practices.
     c) Conduct anonymous quarterly engagement surveys (target: achieve 4.0/5 satisfaction within 18 months).
   - **Expected Impact:** Within a year, raise average job satisfaction from 2.8 to 3.5 and lower total attrition from about 16% to about 10%. enhanced employer brand, increased productivity, and reduced hiring expenses.

3. **Implement a Structured 90-Day Onboarding and Mentorship Program for New Hires**
    - **Rationale:** Based on Insight 1 – employees with <2 years tenure has 28% attrition. Early departures waste recruitment costs (estimated $5,000–$10,000 per employee).
    - **Actionable Steps:**
        a) Create a "new hire survey" at month 3 to identify dissatisfaction early
        b) Create a "new hire survey" at month 3 to identify dissatisfaction early
        c) Assign a senior mentor to every new hire for the first 6 months
    - **Expected Impact:** Reduce first-year attrition by 50% (from 28% to 14%), saving approximately $350,000 annually (based on 100 new hires per year × $7,000 saved per retained employee).

Conclusion

Using the IBM HR Analytics dataset, this Power BI analytical solution effectively tackled the business issue of excessive employee attrition. To facilitate effective reporting, a star schema model was constructed using fact and dimension tables. To monitor important HR parameters, such as attrition rate, job satisfaction, and income benchmarks, eight DAX measures and two computed columns were created. Stakeholders can identify high-risk groups, including early-tenure employees, the sales department, and overtime workers, using the three-page interactive dashboard's executive summary, detailed analysis, and performance monitoring views.

Key insights revealed that employees with less than two years of tenure have a 28% attrition rate, the Sales department accounts for 45% of all attrition, and overtime employees leave at three times the rate of non-overtime staff. Actionable recommendations include implementing a structured onboarding program, launching a Sales retention task force, and introducing company-wide engagement initiatives focused on work-life balance.

For reproducibility, the entire solution—including the Power BI file, transformation documentation, and all screenshots—has been uploaded to GitHub. HR leadership can use this dashboard to make data-driven decisions that lower recruitment costs, increase employee happiness, and decrease turnover.

Part G: GitHub Repository and Documentation

Link: *https://github.com/MMMakena/Business-Intelligence_Sales_Overview/tree/main/Final%20Exam*